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專題名稱	**Anemia Prediction with Deep Learning**				
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摘要

This project asks a simple question: can usable fingernail images carry enough signal for non-invasive anemia screening? Most of the work ended up being the data path before the model, not just the model itself. The pipeline keeps only usable color nail images, crops the nail region from segmentation masks, expands the crop by 1.02x without resizing the original pixels, brightens crops only when they are too dark, and handles reflection highlights conservatively. Small reflections are inpainted, larger ones are softened, and images with too much glare are flagged instead of being over-corrected. After matching images to clinical records, the updated experiments compared six preprocessing variants using the same patient-level split logic. The strongest balanced run used 3,190 nail-crop images from 820 patient records, split by record/patient to avoid image leakage. A frozen MobileNetV3-Small image backbone was fused with age and sex features to predict hemoglobin (HGB) and anemia status. The best setting, safe brightening plus reflection removal, reached test MAE = 1.90 g/dL, Pearson r = 0.52, ROC-AUC = 0.71, anemia precision = 0.76, recall = 0.71, and F1 = 0.73. The result is promising, but still not clinical-grade. I treat it as a proof-of-concept screening pipeline rather than a diagnostic tool.

Keywords: anemia screening; hemoglobin estimation; fingernail images; MobileNetV3; preprocessing; deep learning

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Anemia Prediction with Deep Learning

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Abstract

This project asks a simple question: can usable fingernail images carry enough signal for non-invasive anemia screening? Most of the work ended up being the data path before the model, not just the model itself. The pipeline keeps only usable color nail images, crops the nail region from segmentation masks, expands the crop by 1.02x without resizing the original pixels, brightens crops only when they are too dark, and handles reflection highlights conservatively. Small reflections are inpainted, larger ones are softened, and images with too much glare are flagged instead of being over-corrected. After matching images to clinical records, the updated experiments compared six preprocessing variants using the same patient-level split logic. The strongest balanced run used 3,190 nail-crop images from 820 patient records, split by record/patient to avoid image leakage. A frozen MobileNetV3-Small image backbone was fused with age and sex features to predict hemoglobin (HGB) and anemia status. The best setting, safe brightening plus reflection removal, reached test MAE = 1.90 g/dL, Pearson r = 0.52, ROC-AUC = 0.71, anemia precision = 0.76, recall = 0.71, and F1 = 0.73. The result is promising, but still not clinical-grade. I treat it as a proof-of-concept screening pipeline rather than a diagnostic tool.

***Keywords**:* anemia screening; hemoglobin estimation; fingernail images; MobileNetV3; preprocessing; deep learning

1. Introduction

Anemia is a large public-health problem, and the reviewed papers make it clear why people are interested in image-based screening. Standard hemoglobin testing still depends on blood sampling or specialized equipment, which is not always convenient. Mannino et al. describe anemia as affecting more than 2 billion people and tested smartphone fingernail-bed photos for non-invasive screening [1]. Asare et al. also summarize the practical motivation behind medical-image anemia detection: it can be cheaper, faster, and less invasive than blood-based testing, especially in resource-limited settings [2]. Visible pallor at the conjunctiva, nail beds, tongue, and palms is also discussed in the image-based anemia literature [3,5]. Because phone cameras are already common, fingernail images are a reasonable signal to investigate.

The hard part is that phone nail images are not clean lab images. Lighting changes, cameras change, the finger angle changes, and some crops are either too dark or covered by reflection. If a CNN is trained directly on that kind of data, it can learn shortcuts or fail for reasons that have nothing to do with anemia. Because of that, this paper starts from the bottom of the pipeline: organize the usable images, crop the nail region, control the most obvious lighting problems, then train a lightweight model and check the results.

The goal here is not to replace a complete blood count (CBC). The goal is narrower: build a reproducible pipeline that uses usable nail crops to estimate HGB and screen anemia risk, while avoiding obvious image leakage and avoiding claims that the data cannot support.

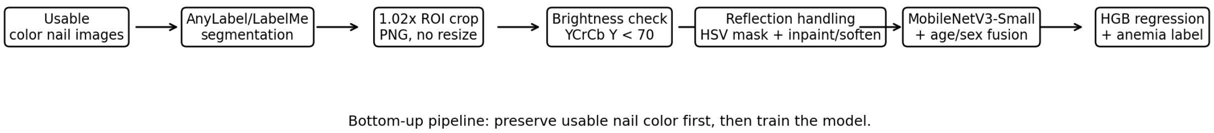

Figure 1. Bottom-up pipeline used in this project. The folder name “glare” in some outputs means brightening plus reflection removal, not a separate clinical feature.

2. Related Work

The reviewed papers point in the same general direction, but they use different body sites and different levels of control. Mannino et al. estimated hemoglobin from smartphone nail-bed photos and image metadata, using CBC hemoglobin as the reference [1]. Asare et al. compared machine-learning models using fingernail, palm, and conjunctiva images, which supports the idea that nail color can contain anemia-related information, although their setup is different from this project [4]. Other studies look at other visible sites: Collings et al. used digital photographs of the conjunctiva [5], Zhang et al. used facial images and deep learning in an emergency-department setting [6], Mitani et al. estimated hemoglobin from retinal fundus images and metadata [3], and Park et al. used smartphone RGB sensing with spectral super-resolution for hemoglobin measurement [7]. The systematic review by Asare et al. is useful background because it shows that image-based anemia detection is an active area, but also that datasets, body sites, validation methods, and clinical assumptions vary a lot [2].

This project should be read in that narrower context. It does not claim to be a finished clinical app, a medical device, or an externally validated system. The contribution is more practical: take an uncontrolled project dataset, keep the usable nail images, make the ROI and preprocessing steps traceable, and test whether a lightweight model still sees a measurable signal. For this kind of project, the pipeline quality matters because the final CNN can only be trusted if the crops, labels, split logic, and preprocessing reports are also trustworthy.

3. Data and Label Construction

The project data comes from three main CSV sources: FileDirectory.csv for image-file metadata, PredictingAnemiaInCo-CJStudyData_DATA_2023-07-12_0653.csv for HGB/HCT/age values, and Race and gender file 070425.csv for demographic fields. This paper uses usable color nail images only. Black-and-white image fields and excluded/unusable image categories are intentionally left outside the scope, so the results describe the usable-image pipeline rather than the full raw dataset.

HGB is used as the regression target. The binary anemia label is derived from HGB and sex: male records are labeled anemia when HGB < 13.0 g/dL, female records when HGB

< 12.0 g/dL, and records with missing or unclear sex use 12.5 g/dL as a fallback. This follows the sex-specific threshold style used in related deep-learning anemia work, while the fallback is only used when sex is missing or unclear [1,6]. Race and ethnicity are not used as model inputs; they are kept only for subgroup reporting.

Table 1. Final patient-level train/validation/test split for the best preprocessing run. Splitting was done by record/patient before image expansion.

Split	Records	Nail crops	Anemia records	Anemia rate	Mean HGB	Mean age
	656	2547	376	57.3%	11.54	59.8
val	82	318	45	54.9%	11.58	59.6
test	82	325	48	58.5%	11.62	55.5

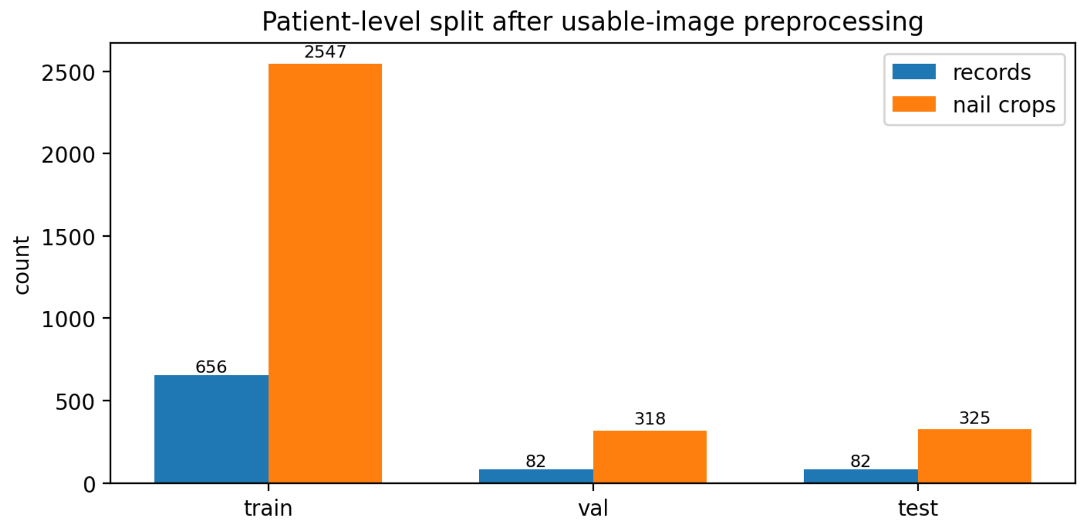

Figure 2. Final patient-level split for the best preprocessing run. Splitting was done by record/patient before image expansion.

4. Preprocessing Pipeline

The preprocessing step was the part where most mistakes could quietly enter the project. The rule I followed was: improve visibility only when needed, and do not change the nail color more than necessary. Since the model is supposed to learn from color-related cues, aggressive enhancement would be risky.

4.1 Nail segmentation and ROI crop

The segmentation notebook reads AnyLabel/LabelMe JSON polygons. The nail region is the target label, and free_edge is treated as an excluded region when it appears. The crop box is expanded directly on the original image coordinates by ROI_SCALE = 1.02. This small detail matters because expanding after a tight crop would not bring back pixels that were already cut away. The crops are saved as PNG to avoid adding another JPEG recompression step.

4.2 Brightening dark crops

Brightness is estimated on the Y channel in YCrCb space while ignoring the black or transparent background. The method is related to low-light enhancement, but this version is deliberately conservative: it only brightens crops that fall below the brightness threshold instead of changing every image [9]. In the attached run, auto mode found 3,467 input crops and produced 3,467 PNG outputs. Only 156 crops were brightened, with zero missing outputs, zero zero-byte outputs, zero shape mismatches, and zero read/processing errors. That report is important because it shows the brightening step was not silently dropping or damaging files.

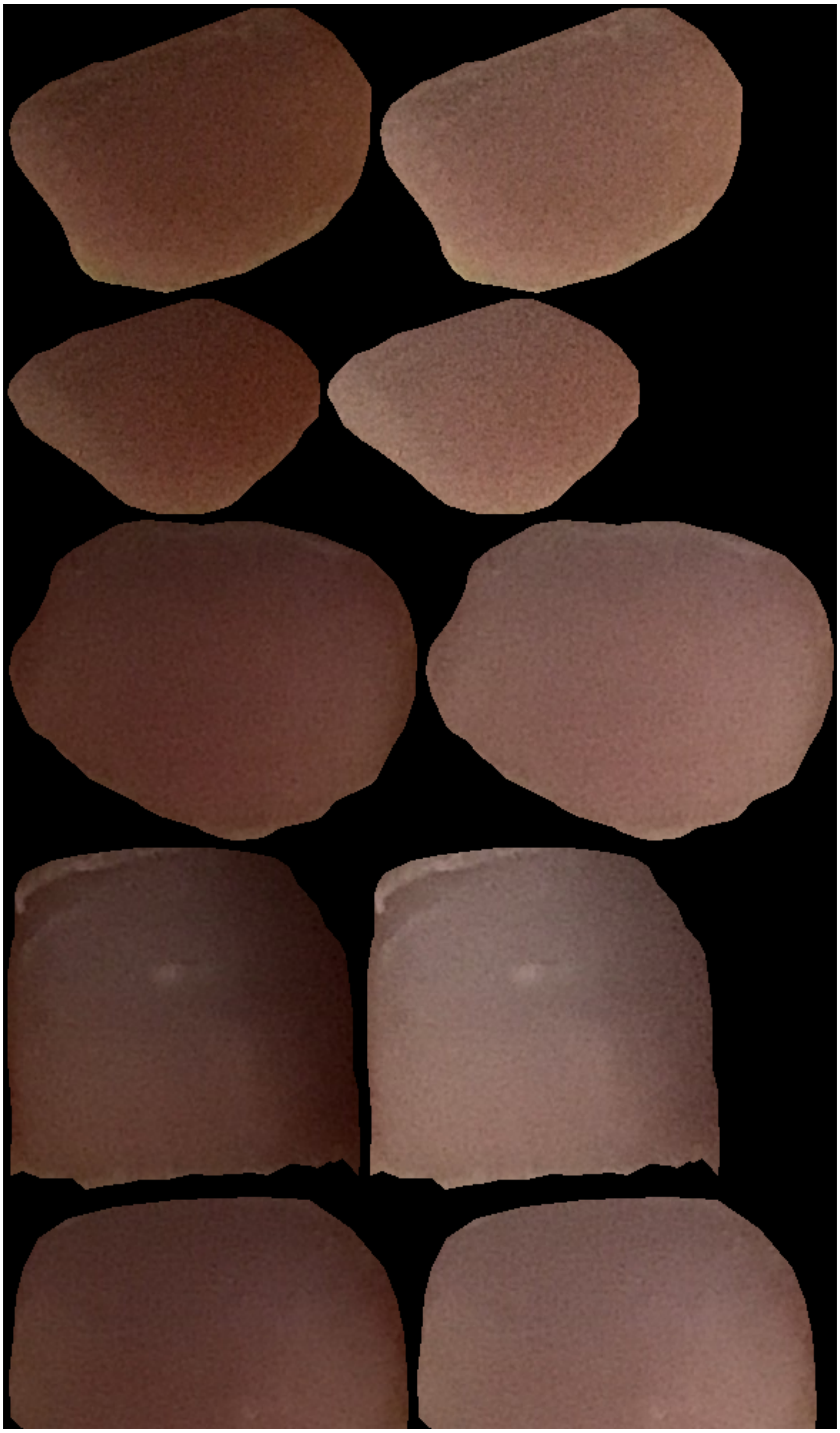

Figure 3. Brightening contact sheet. Left side of each pair is before, right side is after. The goal is visibility, not making every nail look artificially bright.

4.3 Reflection removal / glare handling

In the project folders, the word 'glare' refers to preprocessing, not a medical feature. It means the combination of brightening and reflection handling. Reflection pixels are detected in HSV using a fixed rule (V > 220 and S < 60) and an adaptive rule (V above the 97th percentile of the crop foreground and S < 80). The adaptive rule is useful because some nails are naturally bright, so the mask should focus on the brightest foreground region instead of treating the whole nail as glare. If the reflection area is small (<= 5% of foreground), the pipeline uses inpainting, which is a common image-restoration method for small corrupted regions [10]. If it is medium or large (<= 15%), it softens the reflection. If it is too large (> 15%), it copies the original and marks the crop as unusable_too_much_glare rather than inventing nail color.

Table 2. Preprocessing report summary for the brightening and reflection-removal stages.

Preprocessing report item	Count
Brightening inputs	3467
Brightening outputs	3467
Brightened only because dark	156
Reflection report rows	3467
No glare	1325
Small glare inpainted	1511
Large glare softened	376
Too much glare marked unusable	255

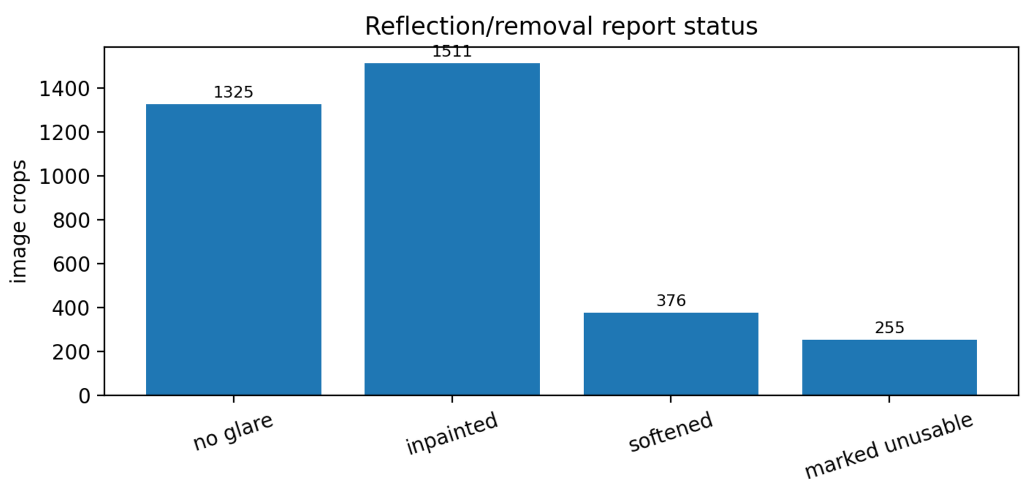

Figure 4. Reflection-removal status counts from reflection_report.csv.

Figure 5. Compact reflection-review example. Red masks mark detected reflection regions.

5. Model and Training Setup

The final training notebook is the MobileNetV3 MacBook-lite v5 version. I kept the anti-leakage logic from the earlier RegNet notebook, but used a smaller model because it was more realistic to train and debug on a MacBook. The image backbone is pretrained MobileNetV3-Small [8] and is frozen by default. Its image feature vector is concatenated with age z-score and sex one-hot encoding. Race and ethnicity are not used as model inputs.

The model has two heads: one for HGB regression and one for anemia probability. The total loss combines HGB regression and anemia classification, with regression weight 1.0 and classification weight 0.5. Images are resized to 128 x 128, the batch size is 4, and patient-balanced sampling is used so that patients with many nail crops do not dominate the training batches.

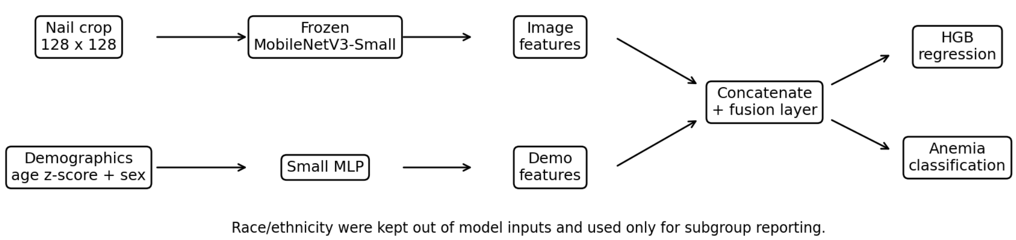

Figure 6. Model structure used in the final MacBook-lite run.

The split is done by record/patient before image expansion. This is necessary here because one patient can produce multiple nail crops. If crops from the same patient appear in both training and testing, the test result can look better than it really is.

6. Results

Table 3. Updated internal comparison of preprocessing settings for HGB regression and anemia classification. “All” keeps every reflection-output image, including copied originals for too-much-glare cases; “safe” excludes images marked unusable_too_much_glare.

Setting	Records	Images	MAE	Pearson r	Bal. Acc.	Precision	Recall	F1	AUC
ROI only	830	3441	2.12	0.39	0.60	0.73	0.40	0.52	0.65
Brightening only	830	3441	2.07	0.42	0.58	0.71	0.36	0.48	0.62
Reflection/all, no bright.	830	3441	2.12	0.39	0.58	0.68	0.40	0.51	0.65
Brightening + reflection/all	830	3441	2.11	0.38	0.57	0.63	0.62	0.62	0.62
Reflection/safe, no bright.	820	3190	1.90	0.51	0.68	0.75	0.69	0.72	0.72
Brightening + reflection/safe	820	3190	1.90	0.52	0.69	0.76	0.71	0.73	0.71

The updated experiment set compared six preprocessing settings. The strongest balanced result came from safe brightening plus reflection removal, where crops marked unusable_too_much_glare were excluded instead of being forced into training. That run used 820 records and 3,190 matched nail crops. Compared with the safe reflection run without brightening, it gave a small regression improvement and the best anemia F1 in the updated experiments. The comparison is still internal to one dataset, so I use it to compare pipeline choices, not to claim clinical validation.

On the final test split, the best run used 82 patient records. HGB regression reached MAE = 1.90 g/dL, RMSE = 2.30 g/dL, R2 = 0.23, Pearson r = 0.52, and Spearman r = 0.50. For anemia classification, the validation-tuned threshold was 0.385. The resulting test scores were accuracy = 0.70, balanced accuracy = 0.69, precision = 0.76, recall = 0.71, F1 = 0.73, ROC-AUC = 0.71, and average precision = 0.80.

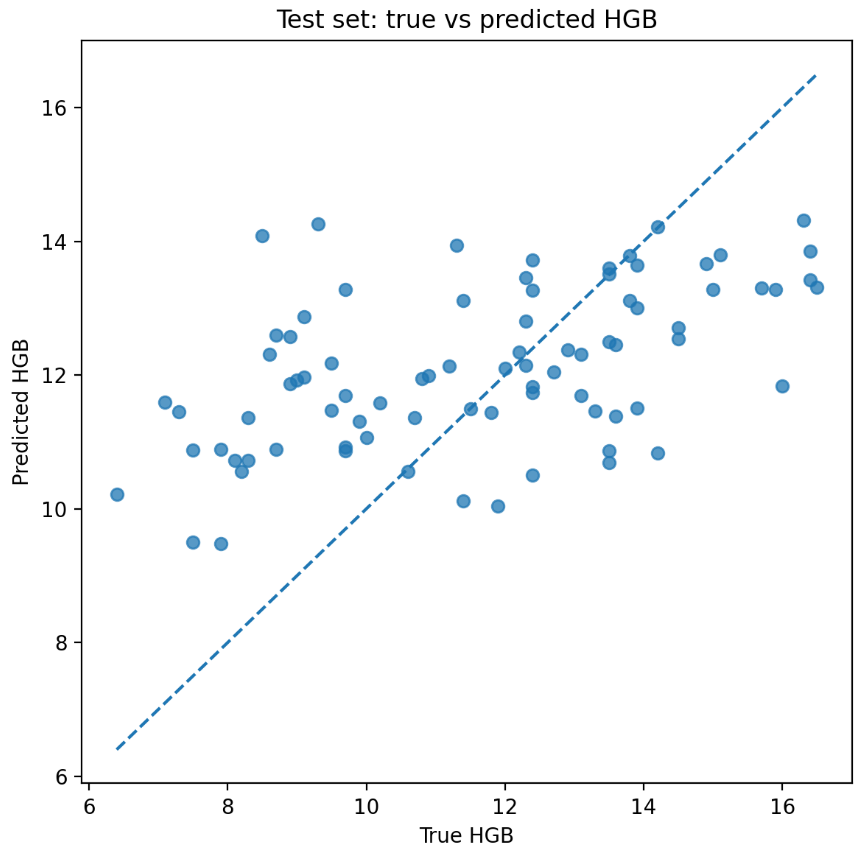

Figure 7. True vs predicted HGB on the test set for the best updated run. The model finds a signal, but predictions are still pulled toward the middle range.

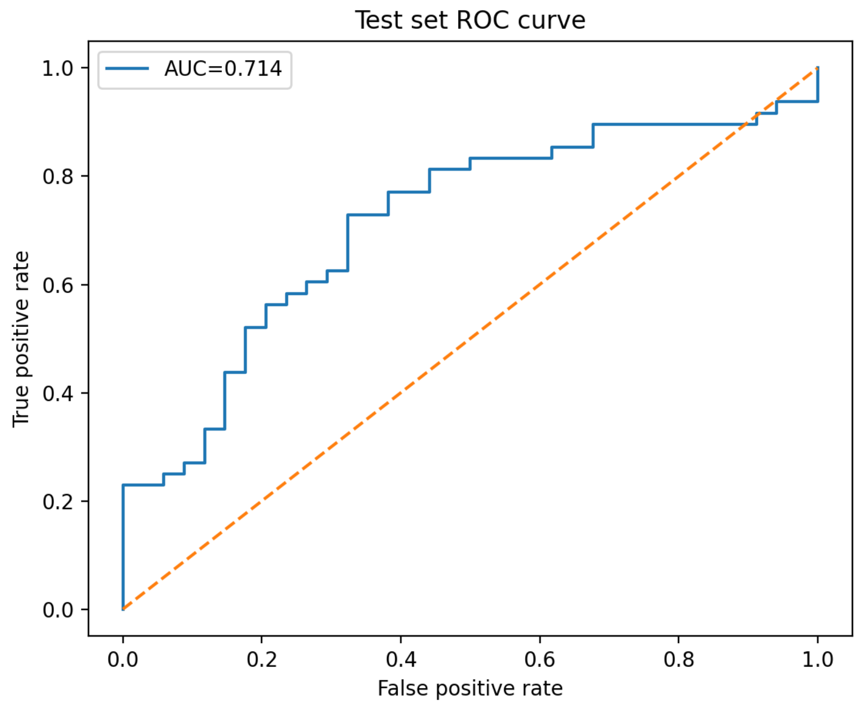

Figure 8. Test ROC curve for anemia classification using the best updated preprocessing setting.

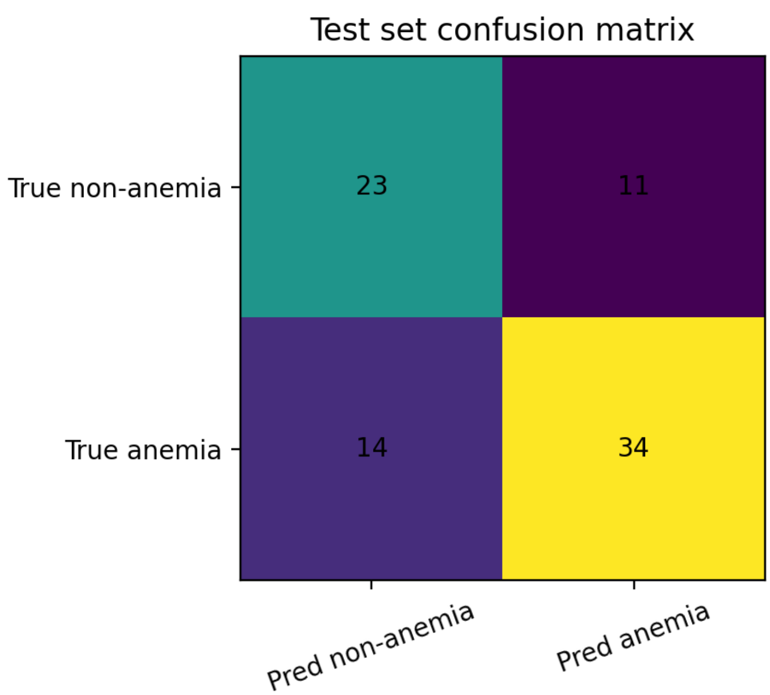

Figure 9. Test confusion matrix. The model detected 34 of 48 anemia records, missed 14, and gave 11 false positives.

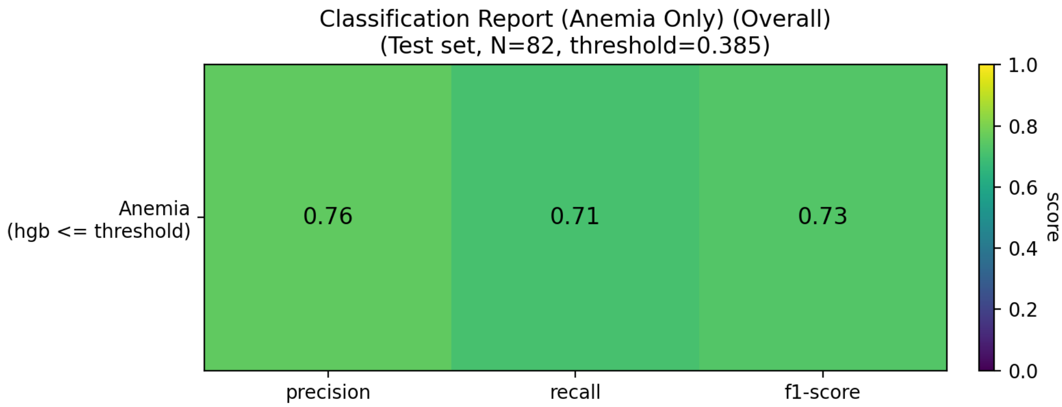

Figure 10. Anemia-only classification report heatmap for the best updated run.

6.1 Subgroup checks

Table 4. Test-set subgroup performance for the best preprocessing run. These values are used as subgroup checks, not final fairness claims, because several groups are small.

Subgroup	N	MAE	Pearson r	Recall	F1
sex_clean: Female	41	1.86	0.38	0.81	0.79
sex_clean: Male	41	1.93	0.52	0.59	0.65
race_group: Asian	5	1.50	0.89	0.50	0.67
race_group: Black	19	1.63	0.71	0.85	0.81
race_group: Hispanic/Latino	9	2.04	0.60	0.80	0.73
race_group: White	47	1.99	0.36	0.67	0.72
ethnicity_clean: HISPANIC	9	2.04	0.60	0.80	0.73
ethnicity_clean: NON-HISPANIC	71	1.89	0.52	0.70	0.74

The subgroup table is included as a check, not as a fairness claim. Some groups are too small to support a strong conclusion; for example, Asian has N=5 and Hispanic/Latino has N=9. The useful takeaway is that subgroup monitoring should be part of the pipeline, but this test set is not large enough to prove fairness or bias mitigation.

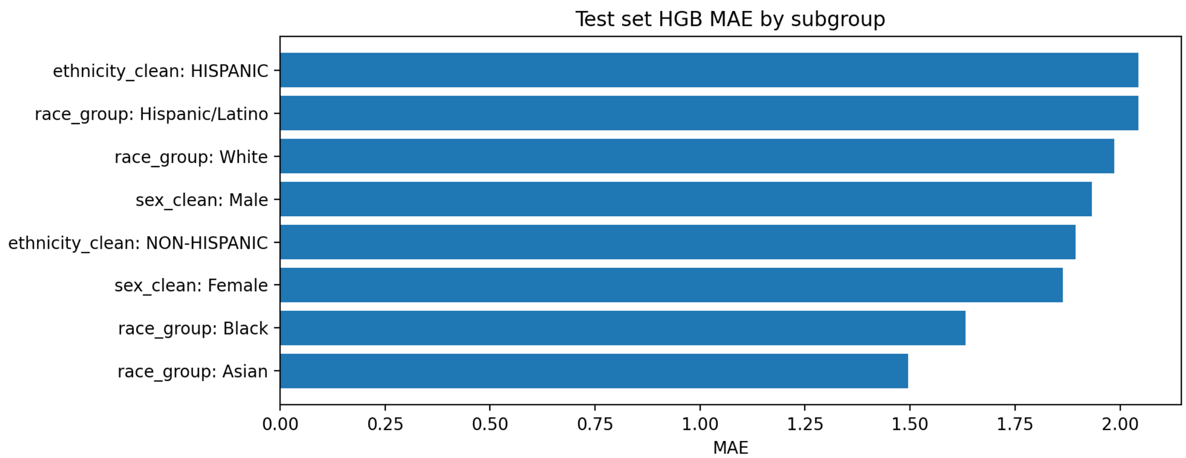

Figure 11. Test MAE by subgroup. This figure is useful for checking problems, not for making final fairness claims.

7. Discussion

The clearest result is that preprocessing mattered, but not in the simple sense that more processing always helped. ROI-only and brightening-only had weaker anemia recall and F1. Keeping all reflection outputs improved recall in one setting, but it also kept copied originals from very high-glare cases. The best overall setting was brightening plus reflection handling after removing crops marked unusable_too_much_glare. This makes sense for this problem: fingernail color and visible pallor can carry useful signal, so dark crops and hard reflections can hide it [1,4,5]. At the same time, color correction can also damage the signal if it is too aggressive. That is why the pipeline stays conservative and records what happened to every crop.

The regression plot also shows a clear weakness. Predictions are pulled toward the middle range: very low HGB values tend to be overestimated, and high HGB values tend to be underestimated. With a limited and imbalanced dataset, this is not surprising, but it matters because some anemia cases are still missed.

The practical use of this system right now is as a research prototype for screening support. A real clinical tool would need external validation, better calibration, more diverse data, locked preprocessing, and a prospectively tested threshold policy. What this project contributes is a traceable pipeline: patient-level splitting, no race/ethnicity input to the model, subgroup checks, and error cases shown directly instead of hidden.

8. Limitations

- No external validation set was available. The test set comes from the same project data source.
- The strongest run has only 82 test records, so subgroup results are unstable.
- Usable-image filtering was done before this paper. The excluded/unusable classifier is not included here.

- The model uses a frozen lightweight backbone so it can be trained and debugged on a MacBook. A larger model or more fine-tuning might improve performance, but it could also overfit without more data.
- Reflection removal is heuristic. It helped this run, but it still needs manual quality review and should not be assumed to work for every dataset.
- The pipeline estimates anemia risk from images and metadata. It does not diagnose anemia or replace CBC testing.

9. Conclusion

This project built a usable-image pipeline for anemia prediction from fingernail crops. The updated experiments compared ROI-only, brightening-only, all-reflection-output, and safe-reflection-output variants. The strongest result came from ROI cropping, selective brightening, and conservative reflection handling while excluding crops marked too much glare, followed by a MobileNetV3-Small image-demographics fusion model. The internal result shows a real signal, with MAE = 1.90 g/dL and anemia F1 = 0.73, but it is not clinical-ready. The most defensible contribution is the pipeline itself: preserve nail quality, prevent leakage, compare preprocessing choices, and report the limits clearly.

10. Reproducibility Checklist

Step	File / output used
Image organization	organize_messy_folder_exact_copy_v2_zero_bytes_fixed.ipynb
Segmentation + ROI	segmentation_roi102_lossless_pipeline_flat_output.ipynb
Brightening	brighten_roi_lossless_pipeline_same_folder.ipynb
Reflection removal	reflection_glare_removal_lossless_pipeline.ipynb
Model training	anemia_mobilenetv3_macbook_lite_REVIEWED_v5_with_update_log.ipynb
Main result folder	safe_usable_nail_crops_roi102_flat_images_brightened_reflection_removed_output_mobilenetv3_lite_outputs/
Main reports	final_record_level_metrics.csv, test_record_level_predictions.csv, test_subgroup_metrics.csv, reflection_report.csv, run_report.json

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